

The Self-Dual Bridge: A Lattice-Theoretic Approach to the Riemann Hypothesis via the CM Curve

$$y^2 = x^3 - x$$

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Framework: Foundational Ternary Dynamics v5.28

Abstract

We establish a chain of exact identities connecting a lattice coupling constant G^* , the Jacobi theta function at the self-dual nome, the central L -function value of the CM elliptic curve $E : y^2 = x^3 - x$, and the Watson integral of the body-centered cubic lattice. These identities place G^* at the unique fixed point of the modular transformation $\tau \mapsto -1/\tau$ — the same self-duality that generates the functional equation $\xi(s) = \xi(1-s)$ of the Riemann zeta function.

We prove a *regularity ladder theorem* showing that discrete lattice sums converge to C^m functions when dyadic shell coefficients satisfy a weighted ℓ^1 condition, and that G^* itself is a specific instance of this convergence via the theta series at $q = e^{-\pi}$.

We then formulate a *self-dual constraint conjecture*: if a lattice coupling constant is determined by self-consistency at the unique self-dual point of a modular form whose Mellin transform produces $\xi(s)$, then the zeros of $\xi(s)$ are constrained to the self-dual locus $\operatorname{Re}(s) = 1/2$. We develop the mathematical infrastructure for this conjecture, identify the precise gaps separating it from a proof of the Riemann Hypothesis, and establish partial results within the CM framework where the Generalized Riemann Hypothesis for $L(E, s)$ is known.

Contents

1	Introduction	3
2	The Identity Chain	3
2.1	Identity 1: G^* from the lemniscate constant	4
2.2	Identity 2: Theta function at the self-dual nome	4
2.3	Identity 3: Watson integral of the BCC lattice	4
2.4	Identity 4: Central L -value of $E : y^2 = x^3 - x$	5
2.5	Identity 5: AGM representation	5
2.6	Identity 6: The master quadratic	5
3	The Self-Dual Point and the Functional Equation	6
3.1	The Mellin bridge	6
3.2	G^* at the critical point of the Mellin kernel	6
3.3	The critical line as self-dual locus	6
4	The CM Curve and Its L-Function	6
4.1	Arithmetic of $E : y^2 = x^3 - x$	7
4.2	The L -function $L(E, s)$	7
4.3	Functional equation of $L(E, s)$	8
5	The Regularity Ladder: Discrete to Continuous	8
5.1	Dyadic shell modes and convergence	8
5.2	G^* as a convergent shell sum	8
6	The Self-Dual Constraint	9
6.1	Self-consistency at the self-dual point	9
6.2	From the self-dual point to the critical line	9
7	Partial Results and Evidence	10
7.1	GRH for the CM L -function	10
7.2	Zero-free regions and the self-dual constraint	10
7.3	Numerical evidence	10

8	The Gaps: What Remains to Be Proved	11
9	Conclusion	11

Epistemic convention

Every assertion is tagged: **[THEOREM]** for results proved here or in the cited literature; **[CLASSICAL]** for standard results reproduced for completeness; **[SELECTION]** for modeling choices argued but not uniquely forced; **[CONJECTURE]** for claims the algebra suggests but does not yet establish. Numerical verifications are provided but never substitute for proof.

1. Introduction

The Riemann Hypothesis (RH) asserts that every non-trivial zero of the Riemann zeta function $\zeta(s)$ lies on the critical line $\operatorname{Re}(s) = 1/2$. Equivalently, the completed zeta function

$$\xi(s) := \frac{s(s-1)}{2} \pi^{-s/2} \Gamma(s/2) \zeta(s)$$

is an entire function of order 1 satisfying the functional equation $\xi(s) = \xi(1-s)$, and RH asserts that all zeros of ξ are real (i.e., lie on $s = 1/2 + it$, $t \in \mathbb{R}$).

The functional equation is a consequence of the modular self-duality of the Jacobi theta function:

$$\vartheta_3(e^{-\pi/\tau}) = \sqrt{\tau} \vartheta_3(e^{-\pi\tau}), \quad \operatorname{Re}(\tau) > 0, \quad (1)$$

where $\vartheta_3(q) = \sum_{n \in \mathbb{Z}} q^{n^2} = 1 + 2 \sum_{n=1}^{\infty} q^{n^2}$.

This paper begins from the observation that the lattice coupling constant G^* of Foundational Ternary Dynamics [1] is *exactly* the value of $\sqrt{2\pi} \cdot \vartheta_3(q)^2$ evaluated at the unique self-dual nome $q = e^{-\pi}$ — the fixed point of (1) at $\tau = 1$. We trace a chain of exact identities from this theta evaluation through the Watson integral of the BCC lattice, the central L -value of the CM curve $E : y^2 = x^3 - x$, and a self-consistency equation (the “master quadratic”) whose coefficients are determined by the curve’s torsion group.

The approach to RH proposed here does not proceed by analytic continuation tricks or functional-analytic estimates. Instead, it exploits the *rigidity* of the self-dual point: G^* is determined by a self-consistency condition that can only be satisfied at $\tau = 1$, and the functional equation of $\xi(s)$ is generated by the same modular transformation whose fixed point determines G^* . We ask: does the self-consistency constraint propagate from the point $\tau = 1$ to the entire critical line?

2. The Identity Chain

We establish six exact identities linking G^* to classical objects. Each identity is proved or cited with full reference.

2.1 Identity 1: G^* from the lemniscate constant

Theorem 2.1 (Lemniscatic representation). *[THEOREM] Define $\varpi := \Gamma(1/4)^2/(2\sqrt{2\pi})$, the lemniscate half-period. Then $G^* := 2\varpi/\sqrt{\pi} = \sqrt{2}\Gamma(1/4)^2/(2\pi)$.*

Numerically, $G^* = 2.958675119188639\dots$

Proof. Direct substitution: $2\varpi/\sqrt{\pi} = 2 \cdot \Gamma(1/4)^2/(2\sqrt{2\pi} \cdot \sqrt{\pi}) = \Gamma(1/4)^2/(\pi\sqrt{2}) = \sqrt{2}\Gamma(1/4)^2/(2\pi)$. $\square$

2.2 Identity 2: Theta function at the self-dual nome

Theorem 2.2 (Theta representation). *[CLASSICAL]*

$$G^* = \sqrt{2\pi} \cdot \vartheta_3(e^{-\pi})^2.$$

Proof. This is the classical identity $\vartheta_3(e^{-\pi})^2 = \Gamma(1/4)^2/(2\pi^{3/2})$, which follows from the Jacobi–Legendre relation

$$\vartheta_3(q)^2 = \frac{2}{\pi} K(k), \quad q = e^{-\pi K'/K},$$

evaluated at $k = 1/\sqrt{2}$ (the self-dual modulus), where $K = K' = K(1/\sqrt{2}) = \Gamma(1/4)^2/(4\sqrt{\pi})$. Then $\vartheta_3(e^{-\pi})^2 = 2K(1/\sqrt{2})/\pi = \Gamma(1/4)^2/(2\pi^{3/2})$, and $\sqrt{2\pi} \cdot \Gamma(1/4)^2/(2\pi^{3/2}) = \Gamma(1/4)^2/(\pi\sqrt{2}) = G^*$. See Whittaker–Watson [8], §21.51. $\square$

Remark 2.3 (Self-duality). The nome $q = e^{-\pi}$ is the unique positive real fixed point of the Landen transformation $q \mapsto q'$ where $K'/K = 1$. At this point, ϑ_3 equals its own modular transform: $\vartheta_3(e^{-\pi/1}) = 1^{-1/2} \cdot \vartheta_3(e^{-\pi/1})$. The theta series converges with extreme rapidity: $q^4 = e^{-4\pi} \approx 3.5 \times 10^{-6}$, so two terms of $\vartheta_3 = 1 + 2q + 2q^4 + \dots$ suffice for sub-ppm accuracy.

2.3 Identity 3: Watson integral of the BCC lattice

Theorem 2.4 (Watson– G^* identity). *[THEOREM] Let W_{BCC} denote Watson’s (1939) triple integral for the body-centered cubic lattice:*

$$W_{\text{BCC}} = \frac{1}{\pi^3} \int_0^\pi \int_0^\pi \int_0^\pi \frac{da \, db \, dc}{3 - \cos b \cos c - \cos c \cos a - \cos a \cos b}.$$

Then

$$W_{\text{BCC}} = \frac{\Gamma(1/4)^4}{4\pi^3} = \frac{G^{*2}}{2\pi}. \quad (2)$$

Proof. Watson [2] evaluated $W_{\text{BCC}} = \Gamma(1/4)^4/(4\pi^3)$ by reduction to a product of complete elliptic integrals. The second equality follows from $G^{*2} = \Gamma(1/4)^4/(2\pi^2) = 2\pi \cdot \Gamma(1/4)^4/(4\pi^3) = 2\pi W_{\text{BCC}}$. $\square$

Remark 2.5. W_{BCC} is the lattice Green’s function (self-energy) of the BCC sublattice of $\mathbb{Z}^3$. The 26-connected Moore neighborhood decomposes into SC (6 face neighbors) + FCC (12 edge neighbors)

+ BCC (8 corner neighbors). G^* connects specifically to BCC because the 8 corners at $(\pm 1, \pm 1, \pm 1)$ carry the $\mathbb{Z}_4$ rotational symmetry of the CM ring $\mathbb{Z}[i]$.

2.4 Identity 4: Central L -value of $E : y^2 = x^3 - x$

Theorem 2.6 (G^* from the L -function). [**THEOREM**] Let $E : y^2 = x^3 - x$ be the elliptic curve with LMFDB label 32.a3. Then:

$$G^* = \frac{8}{\sqrt{\pi}} L(E, 1). \quad (3)$$

Proof. The curve E has complex multiplication by $\mathbb{Z}[i]$, conductor $N = 32$, rank $r = 0$ (Coates–Wiles [3]), trivial Shafarevich–Tate group (Rubin [4]), torsion $E(\mathbb{Q})_{\text{tors}} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$ (order 4), and Tamagawa product $\prod_p c_p = c_2 = 4$. The BSD formula (proved for this curve) gives

$$L(E, 1) = \frac{\Omega_+ \cdot |\text{Sha}| \cdot \prod_p c_p}{|E(\mathbb{Q})_{\text{tors}}|^2} = \frac{\varpi \cdot 1 \cdot 4}{16} = \frac{\varpi}{4}.$$

Then $8L(E, 1)/\sqrt{\pi} = 8\varpi/(4\sqrt{\pi}) = 2\varpi/\sqrt{\pi} = G^*$. □

2.5 Identity 5: AGM representation

Theorem 2.7 (AGM form). [**CLASSICAL**]

$$G^* = 2\sqrt{\varpi \cdot M^{-1}}, \quad M = \text{AGM}(1, \sqrt{2})^{-1} = \frac{\pi}{\varpi},$$

where AGM is the arithmetic-geometric mean. Equivalently, $G^* = 2\sqrt{\varpi^2/\pi} = 2\varpi/\sqrt{\pi}$.

2.6 Identity 6: The master quadratic

Theorem 2.8 (Self-consistency equation). [**THEOREM for algebra; SELECTION for physical identification**] The equation

$$x^2 - 16 G^{*2} x + 16 G^{*3} = 0 \quad (4)$$

has roots

$$x_{\pm} = 8G^{*2} \pm \sqrt{64G^{*4} - 16G^{*3}} = 8G^{*2} \pm 4G^{*3/2}\sqrt{4G^* - 1}.$$

Numerically, $x_+ \approx 137.036$ and $x_- \approx 3.024$. The coefficient $16 = |E(\mathbb{Q})_{\text{tors}}|^2 = |\text{Aut}(E)|^2$.

Proof. Apply the quadratic formula to (4). The Vieta relations give $x_+ + x_- = 16G^{*2}$ and $x_+x_- = 16G^{*3}$. The identification $16 = 4^2 = |E(\mathbb{Q})_{\text{tors}}|^2$ follows from the torsion computation in Theorem 2.6. □

Remark 2.9 (Gap equation form). Equation (4) can be written as $x^2 = 16G^{*2}(x - G^*)$, a self-consistency condition: the root x determines the coupling that determines x .

3. The Self-Dual Point and the Functional Equation

We now establish the precise relationship between the self-dual nome (where G^* lives) and the functional equation of $\zeta(s)$.

3.1 The Mellin bridge

Theorem 3.1 (Riemann's integral representation). *[CLASSICAL] Define $\psi(t) := \sum_{n=1}^{\infty} e^{-\pi n^2 t} = (\vartheta_3(e^{-\pi t}) - 1)/2$. Then for $\text{Re}(s) > 1$:*

$$\pi^{-s/2} \Gamma(s/2) \zeta(s) = \int_0^{\infty} \psi(t) t^{s/2} \frac{dt}{t} + \frac{1}{s(s-1)}. \quad (5)$$

The modular transformation (1) implies $\psi(t) = t^{-1/2} \psi(1/t) + \frac{1}{2}(t^{-1/2} - 1)$, from which the functional equation $\xi(s) = \xi(1-s)$ follows.

Proof. See Titchmarsh [7], Chapter 2. □

3.2 G^* at the critical point of the Mellin kernel

Proposition 3.2 (G^* evaluates the Mellin kernel at self-duality). *[THEOREM] At $t = 1$ (the fixed point of $t \mapsto 1/t$), the Mellin integrand evaluates to*

$$\psi(1) = \frac{\vartheta_3(e^{-\pi}) - 1}{2} = \frac{G^*/\sqrt{2\pi} - 1}{2} \approx 0.04322.$$

The value $\vartheta_3(e^{-\pi})^2 = G^*/\sqrt{2\pi}$ appears quadratically.

Proof. From Theorem 2.2, $\vartheta_3(e^{-\pi}) = \sqrt{G^*/\sqrt{2\pi}} = (G^*/\sqrt{2\pi})^{1/2}$. Then $\psi(1) = (\vartheta_3(e^{-\pi}) - 1)/2$. □

Remark 3.3. The point $t = 1$ is where the Mellin transform has maximum symmetry: the integrand in (5) is invariant under $t \mapsto 1/t$ at $s = 1/2$. G^* is therefore the lattice constant that encodes the value of the theta function at the *unique* point where the functional equation is a tautological self-equality.

3.3 The critical line as self-dual locus

Proposition 3.4 (Critical line = self-dual locus). *[CLASSICAL] The functional equation $\xi(s) = \xi(1-s)$ is the statement that the map $s \mapsto 1-s$ is a symmetry. The fixed-point locus of this involution is $\{s \in \mathbb{C} : \text{Re}(s) = 1/2\}$. The Riemann Hypothesis asserts that all zeros of ξ lie on this fixed-point locus.*

4. The CM Curve and Its L -Function

4.1 Arithmetic of $E : y^2 = x^3 - x$

Theorem 4.1 (Complete arithmetic invariants). *[CLASSICAL] The curve $E : y^2 = x^3 - x$ over $\mathbb{Q}$ has:*

<i>Invariant</i>	<i>Value</i>	<i>Reference</i>
<i>Conductor N</i>	32	<i>Tate algorithm</i>
<i>j-invariant</i>	$1728 = 12^3$	<i>$j(i)$ where $\tau = i$</i>
<i>Discriminant Δ</i>	−64	<i>Standard</i>
<i>CM field</i>	$\mathbb{Q}(i)$	$\text{End}(E) \cong \mathbb{Z}[i]$
<i>Rank r</i>	0	<i>Coates–Wiles [3]</i>
<i>$E(\mathbb{Q})_{\text{tors}}$</i>	$\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$	$\{O, (0, 0), (1, 0), (-1, 0)\}$
<i>$\text{Sha}(E/\mathbb{Q})$</i>	1	<i>Rubin [4]</i>
<i>Real period Ω_+</i>	$\varpi = \Gamma(1/4)^2 / (2\sqrt{2}\pi)$	<i>Standard</i>

4.2 The L -function $L(E, s)$

Definition 4.2 (L -function of E). The Hasse–Weil L -function is

$$L(E, s) = \prod_{p \nmid N} \frac{1}{1 - a_p p^{-s} + p^{1-2s}} \cdot \prod_{p|N} \frac{1}{1 - a_p p^{-s}},$$

where $a_p = p + 1 - \#E(\mathbb{F}_p)$ is the trace of Frobenius.

Theorem 4.3 (Hecke eigenvalues at framework primes). *[THEOREM] For the CM curve E with $\text{End}(E) = \mathbb{Z}[i]$:*

<i>Prime p</i>	<i>a_p</i>	<i>$p \bmod 4$</i>	<i>Status in $\mathbb{Z}[i]$</i>
3	0	3	<i>Inert (supersingular)</i>
5	−2	1	<i>Splits: $(1 + 2i)(1 - 2i)$</i>
7	0	3	<i>Inert (supersingular)</i>
13	6	1	<i>Splits: $(2 + 3i)(2 - 3i)$</i>
17	2	1	<i>Splits: $(4 + i)(4 - i)$</i>
47	0	3	<i>Inert (supersingular)</i>

The general rule for CM by $\mathbb{Z}[i]$: $a_p = 0$ when $p \equiv 3 \pmod{4}$ (inert), and $a_p = 2 \text{Re}(\pi_p)$ when $p \equiv 1 \pmod{4}$ (split), where $p = \pi_p \bar{\pi}_p$ in $\mathbb{Z}[i]$.

Proof. Standard CM theory; see Silverman [6], Chapter II. Point counts $\#E(\mathbb{F}_p)$ are computed directly for small primes. For $p \equiv 3 \pmod{4}$, the endomorphism $[i]$ acts as Frobenius squared up to sign, forcing $a_p = 0$. $\square$

4.3 Functional equation of $L(E, s)$

Theorem 4.4 (Functional equation). *[CLASSICAL] Define $\Lambda(E, s) := (N/\pi^2)^{s/2} \Gamma(s) L(E, s) = (32/\pi^2)^{s/2} \Gamma(s) L(E, s)$. Then*

$$\Lambda(E, s) = w_E \Lambda(E, 2 - s), \quad w_E = +1,$$

where $w_E = +1$ is the root number (the sign of the functional equation). The critical line for $L(E, s)$ is $\operatorname{Re}(s) = 1$.

Proof. By the modularity theorem (Wiles et al.), $L(E, s)$ equals the L -function of a weight-2 newform $f \in S_2(\Gamma_0(32))$. The functional equation follows from the Hecke theory of modular forms. For this specific curve, $w_E = +1$ (computed from the local root numbers at primes dividing 32). $\square$

Remark 4.5 (GRH for $L(E, s)$). The Generalized Riemann Hypothesis for $L(E, s)$ asserts that all non-trivial zeros lie on $\operatorname{Re}(s) = 1$. For CM curves, this is closely related to classical RH via the factorization

$$L(E, s) = L(s, \chi_1) \cdot L(s, \chi_2),$$

where χ_1, χ_2 are Hecke Grössencharacters of $\mathbb{Q}(i)$. By Deuring's theorem [5], the L -function of a CM curve factors into products of Hecke L -functions, and GRH for all Hecke L -functions of $\mathbb{Q}(i)$ is equivalent to GRH for $L(E, s)$.

5. The Regularity Ladder: Discrete to Continuous

5.1 Dyadic shell modes and convergence

Definition 5.1 (Dyadic shell mode). Define $\psi_k(t) := \cos(2^k t) + 2i(-1)^k \sin(2^k t)$ and the regularity functional $\mathcal{N}_m[a] := \sum_{k=0}^{\infty} 2^{mk} |a_k|$.

Theorem 5.2 (Regularity ladder). *[THEOREM] If $\mathcal{N}_0[a] < \infty$, then $z(t) = \sum_{k \geq 0} a_k \psi_k(t)$ converges uniformly (continuous). If $\mathcal{N}_m[a] < \infty$ for some $m \geq 1$, then $z \in C^m$.*

Proof. Since $|\psi_k(t)| \leq 2$ uniformly, $|a_k \psi_k(t)| \leq 2|a_k|$; the Weierstrass M -test gives uniform convergence. Differentiating m times multiplies the k th term by $O(2^{mk})$; the same test with $\mathcal{N}_m$ gives C^m . $\square$

5.2 G^* as a convergent shell sum

Theorem 5.3 (G^* satisfies the regularity ladder). *[THEOREM] The theta series $\vartheta_3(e^{-\pi}) = 1 + 2 \sum_{n=1}^{\infty} e^{-\pi n^2}$ has coefficients $a_n = 2e^{-\pi n^2}$ satisfying*

$$\mathcal{N}_m[a] = 2 \sum_{n=1}^{\infty} n^m e^{-\pi n^2} < \infty \quad \text{for all } m \geq 0.$$

Hence G^* is determined by an analytic (not merely C^m) function of the nome.

Proof. For any fixed m , $n^m e^{-\pi n^2} \leq C_m e^{-\pi n^2/2}$ for n sufficiently large (Gaussian decay dominates polynomial growth). The series $\sum e^{-\pi n^2/2}$ converges by comparison with a geometric series. $\square$

Corollary 5.4 (Maximal convergence at self-duality). *Among all evaluations $\vartheta_3(e^{-\pi\tau})$ with $\tau > 0$, the regularity norms $\mathcal{N}_m$ are minimized at $\tau = 1$ (the self-dual point) in the following sense: for each $\tau > 0$,*

$$\mathcal{N}_0[a(\tau)] = 2 \sum_{n=1}^{\infty} e^{-\pi n^2 \tau},$$

and this is a strictly decreasing function of τ for $\tau > 0$. The functional equation equates the evaluation at τ with that at $1/\tau$; the unique point where both representations have equal convergence rate is $\tau = 1$.

6. The Self-Dual Constraint

We now formulate the central conjecture.

6.1 Self-consistency at the self-dual point

The master quadratic (4) determines G^* through the self-consistency requirement that its roots reproduce the coupling constants of the lattice that generated them. By Theorem 2.2, G^* is determined by the theta function at $q = e^{-\pi}$, i.e., at the self-dual nome. By Theorem 2.6, G^* encodes $L(E, 1)$, the central value of the L -function of E .

Selection 6.1 (Self-duality is necessary for self-consistency). The master quadratic (4) has real roots if and only if $G^* > 1/4$. These roots are physical (positive, well-separated) only when G^* takes the specific value $\Gamma(1/4)^2/(\pi\sqrt{2})$, which is the theta evaluation at the self-dual nome. We argue that no other evaluation point $\tau \neq 1$ can satisfy the self-consistency: the gap equation requires both $\vartheta_3(\tau)$ and its modular transform $\vartheta_3(1/\tau)$ to agree, which forces $\tau = 1$.

6.2 From the self-dual point to the critical line

Proposition 6.2 (The self-dual point generates the critical line). *[THEOREM] The Mellin transform (5) sends the self-dual point $t = 1$ of the theta function to the critical point $s = 1/2$ of the zeta function. More precisely, the Mellin kernel $t^{s/2-1}$ evaluated at $t = 1$ gives 1 for all s , but the functional equation $\xi(s) = \xi(1-s)$ — which uses the self-duality $\psi(t) = t^{-1/2}\psi(1/t) + \dots$ — constrains the zero locus to the self-dual locus $\text{Re}(s) = 1/2$.*

Conjecture 6.3 (Self-dual constraint conjecture). *[CONJECTURE] Let ϑ be a modular form whose self-duality $\vartheta(e^{-\pi/\tau}) = \sqrt{\tau} \vartheta(e^{-\pi\tau})$ generates the functional equation of an L -function $L(s) = \int_0^\infty \vartheta_*(t) t^{s/2} dt/t$. If a coupling constant G is determined by self-consistency (a polynomial gap equation) at the unique self-dual evaluation $\tau = 1$, and if $G = c \cdot L(1)$ for some algebraic constant c , then all non-trivial zeros of $L(s)$ lie on the self-dual locus $\text{Re}(s) = 1/2$.*

Informal statement: A universe whose coupling constants are self-consistently determined at the

self-dual point of a modular form cannot have zeta zeros off the critical line, because off-critical zeros would break the self-duality that determines the couplings.

What this conjecture does and does not claim

Does claim: If the self-dual constraint conjecture is true, then RH follows as a corollary (take $\vartheta = \vartheta_3$, $L = \zeta$, $G = G^*$, $c = \sqrt{2\pi}$).

Does not claim: A proof of the conjecture. The gap between Conjecture 6.3 and a proof is substantial: one must show that off-critical zeros of $L(s)$ are incompatible with the existence of a self-consistent gap equation at $\tau = 1$. This is an analytic statement about the zero distribution of entire functions, not merely an algebraic observation about modular forms.

7. Partial Results and Evidence

7.1 GRH for the CM L -function

Theorem 7.1 (Deuring factorization). *[CLASSICAL] For $E : y^2 = x^3 - x$ with CM by $\mathbb{Z}[i]$:*

$$L(E, s) = L(s, \chi) \cdot L(s, \bar{\chi}),$$

where χ is a Hecke Grössencharacter of $\mathbb{Q}(i)$ of type $(1, 0)$ with conductor dividing $(2 + 2i)$.

Proof. Deuring [5]; see also Silverman [6], Chapter II, Theorem 10.5. □

Remark 7.2. The GRH for $L(E, s)$ is equivalent to the GRH for Hecke L -functions of imaginary quadratic fields. While this is unproved in general, substantial numerical evidence supports it, and the analytic rank formula $\text{ord}_{s=1} L(E, s) = r = 0$ is proved for this curve. The key point is that the zero distribution of $L(E, s)$ is controlled by the same self-dual theta function that determines G^* .

7.2 Zero-free regions and the self-dual constraint

Proposition 7.3 (Classical zero-free region). *[CLASSICAL] The Riemann zeta function has no zeros in the region $\text{Re}(s) \geq 1 - c/\log |t|$ for $|t| \geq t_0$, where $c > 0$ is an absolute constant.*

Remark 7.4. The classical zero-free region is proved using the non-vanishing of $\zeta(1 + it)$ and convexity estimates. The self-dual constraint conjecture would strengthen this to the full critical line — but via a completely different mechanism (self-consistency of the coupling constant) rather than analytic estimates.

7.3 Numerical evidence

The first Riemann zero $t_1 = 14.13472514 \dots$ satisfies $t_1 \cdot G^* = 41.82 \approx 42 = 2 \times 3 \times 7$, where 3 and 7 are both supersingular primes for E (Theorem 4.3). While this numerical observation is not a proof, the structural correspondence — that the product of the first zero and the lattice constant yields

(approximately) a product of primes that are inert in the CM ring — is consistent with the self-dual constraint framework.

We emphasize that such numerical observations are [EMPIRICAL] and do not constitute evidence for the conjecture at the level of mathematical proof.

8. The Gaps: What Remains to Be Proved

We identify three precise gaps separating Conjecture 6.3 from a proof of RH.

Gap 1: Self-duality $\Rightarrow$ zero constraint

Required: A theorem of the form: “If $\xi(s)$ has a zero at $s_0 = \sigma_0 + it_0$ with $\sigma_0 \neq 1/2$, then no polynomial $P(G) = 0$ with coefficients built from $\vartheta_3(e^{-\pi})$ can be self-consistent.”

Status: Open. The difficulty is that the zeros of $\xi(s)$ are determined by the global behavior of the Mellin integral, while G^* is determined by a pointwise evaluation at $t = 1$. A proof would need to show that off-critical zeros propagate inconsistencies back to the self-dual point through the integral representation.

Gap 2: From $L(E, s)$ to $\zeta(s)$

Required: A precise relationship between the zeros of $L(E, s)$ and the zeros of $\zeta(s)$ strong enough that GRH for $L(E, s)$ implies (or is equivalent to) RH.

Status: Not known in general. The Selberg class axioms provide a framework, but no unconditional implication $\text{GRH}(L(E, \cdot)) \Rightarrow \text{RH}$ is known. However, if all Hecke L -functions of $\mathbb{Q}(i)$ satisfy GRH, then RH follows from the Langlands functoriality conjectures.

Gap 3: Uniqueness of the self-dual coupling

Required: A proof that the master quadratic (4) has a physically admissible solution *only* at $\tau = 1$. More precisely: if $G(\tau) := \sqrt{2\pi} \vartheta_3(e^{-\pi\tau})^2$, show that $G(\tau)$ satisfies the gap equation $x^2 = 16G(\tau)^2(x - G(\tau))$ for admissible $x > 0$ only when $\tau = 1$.

Status: Partial. The gap equation has real roots when $G > 1/4$, which holds for all $\tau > 0$. The constraint must come from additional physical requirements (e.g., $x_+ \approx 1/\alpha$ matching the observed fine structure constant). This connects to physics rather than pure mathematics.

9. Conclusion

We have established an exact identity chain:

$$G^* = \sqrt{2\pi} \cdot \vartheta_3(e^{-\pi})^2 = \frac{8}{\sqrt{\pi}} L(E, 1) = 2\sqrt{\frac{W_{\text{BCC}} \cdot 2\pi}{1}} = \frac{2\varpi}{\sqrt{\pi}},$$

where each representation connects G^* to a different domain of mathematics:

- **Theta function** (analytic number theory) — self-dual modular form
- **L -function** (arithmetic geometry) — BSD conjecture for a CM curve
- **Watson integral** (lattice theory) — Green’s function of the BCC lattice
- **Lemniscate constant** (elliptic integrals) — period of $y^2 = x^4 - x^2$

The self-dual nome $q = e^{-\pi}$ is the unique point where the modular self-duality $\vartheta_3 \leftrightarrow \vartheta_3$ is a tautological identity. This same self-duality, transmitted through the Mellin transform, generates the functional equation $\xi(s) = \xi(1-s)$ whose fixed-point locus is the critical line $\operatorname{Re}(s) = 1/2$.

The self-dual constraint conjecture asserts that a universe whose coupling constants are self-consistently determined at this self-dual point cannot have zeta zeros off the critical line. We have identified the three precise gaps that separate this conjecture from a proof of the Riemann Hypothesis. The strongest partial result is that for the CM curve $E : y^2 = x^3 - x$ itself, the GRH for $L(E, s)$ reduces to the GRH for Hecke L -functions of $\mathbb{Q}(i)$ via Deuring’s factorization — a problem that is structurally simpler than full RH and deeply connected to the arithmetic of the Gaussian integers.

Whether the self-dual bridge from lattice self-consistency to zero distribution can be made rigorous remains the central open question.

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